Ps 04

Erik An

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1 Problem 1 DONE

1.1 Task

Verify the following trigonometric identities and check them in Julia. $\cos(-x) = \cos(x)$ $\sin(-x) = -\sin(x)$ $\cos(x) = \sin(x + \frac{\pi}{2})$ $\sin(x) = \cos(x - \frac{\pi}{2})$ $\sin(x + \pi) = -\sin(x)$ $\cos(x + \pi) = -\cos(x)$,

1.2 Solution

1.2.1 $\cos(-x) = \cos(x)$

Tailor Series Expansion

$$cos(x) = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!} \qquad sin(x) = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!}$$

Lets assume $cos(-x) = cos(x)$. Hence,
$\sum_{k=0}^{\infty} \frac{(-1)^k (-x)^{2k}}{(2k)!} = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!}$
Since $k \in \mathbb{N}_0$, $(-x)^{2k} = x^{2k}$. Thus,
$\sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!} = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!}$
$cos(-x) = cos(x)$

1.2.2 $\sin(-x) = -\sin(x)$

Tailor Series Expansion

$$cos(x) = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!} \qquad sin(x) = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!}$$

$sin(-x) = \sum_{k=0}^{\infty} \frac{(-1)^k (-x)^{2k+1}}{(2k+1)!}$
Expression $2k+1$, where $k \in \mathbb{N}_0$ would always be odd.
$sin(-x) = -\sum_{k=0}^{\infty} \frac{(-1)^k (x)^{2k+1}}{(2k+1)!}$
$sin(-x) = -sin(x)$

1.2.3 $\cos(x) = \sin(x + \frac{\pi}{2})$

Trigonometric identity:

$$\sin(A + B) = \sin(A)\cos(B) + \cos(A)\sin(B)$$

$$sin(x + \frac{\pi}{2}) = sin(x)cos(\frac{\pi}{2}) + cos(x)sin(\frac{\pi}{2})$$
$$sin(x + \frac{\pi}{2}) = cos(x)$$

1.2.4 $\sin(x) = \cos(x - \frac{\pi}{2})$

Trigonometric identity:

$$\cos(A - B) = cos(A)cos(B) + sin(A)sin(B)$$

$$\cos(x - \frac{\pi}{2}) = cos(x)cos(\frac{\pi}{2}) + sin(x)sin(\frac{\pi}{2})$$
$$\cos(x - \frac{\pi}{2}) = sin(x)$$

1.2.5 $\sin(x + \pi) = -\sin(x)$

Trigonometric identity:

$$\sin(A + B) = \sin(A)\cos(B) + \cos(A)\sin(B)$$

$$\sin(x + \pi) = \sin(x)\cos(\pi) + \cos(x)\sin(\pi)$$
$$\sin(x + \pi) = -\sin(x)$$

1.2.6 $\cos(x + \pi) = -\cos(x)$

Trigonometric identity:

$$\cos(A + B) = cos(A)cos(B) - sin(A)sin(B)$$

$$\cos(x + \pi) = cos(x)cos(\pi) - sin(x)sin(\pi)$$
$$\cos(x + \pi) = -cos(x)$$

1.2.7 code

```
using Symbolics

euler_cos() = (exp(im*) + exp(-im*)) / 2
euler_sin() = (exp(im*) - exp(-im*)) / (2im)

function validate(label, LHS, RHS)
    # symbolic difference
    diff = LHS - RHS

    sym_diff = simplify(diff)

    @variables x
    num_diff = substitute(sym_diff, Dict(x => 1.0))

    val = Symbolics.unwrap(num_diff)

    if abs(val) < 1e-12
        println(rpad(label, 30), "  Q.E.D.")
    else
        println(rpad(label, 30), "  Not proven. Diff  $val")
    end
end

@variables x

println("--- PROOFS (Euler + Numerical Check) ---")
validate("cos(-x) = cos(x)",       euler_cos(-x), euler_cos(x))
validate("sin(-x) = -sin(x)",      euler_sin(-x), -euler_sin(x))
validate("cos(x) = sin(x + /2)",  euler_cos(x),   euler_sin(x + pi/2))
validate("sin(x) = cos(x - /2)",  euler_sin(x),   euler_cos(x - pi/2))
validate("sin(x + ) = -sin(x)",   euler_sin(x+pi), -euler_sin(x))
validate("cos(x + ) = -cos(x)",   euler_cos(x+pi), -euler_cos(x))
```

2 Problem 2 DONE

2.1 Task

Determine the derivative of arccos and check in Julia.

2.2 Solution

2.2.1 Derivation

$y = cos^{-1}(x)$

$$cos(y) = x$$
$$\frac{d}{dx}[cos(y)] = \frac{d}{dx}[x]$$
$$-sin(y)\frac{dy}{dx} = 1$$
$$\frac{dy}{dx} = -\frac{1}{sin(y)}$$
$$sin(y)^2 + cos(y)^2 = 1$$
$$sin(y)^2 + x^2 = 1$$
$$sin(y) = \sqrt{1 - x^2}$$
$$\frac{dy}{dx} = -\frac{1}{\sqrt{1 - x^2}}$$

2.2.2 Code

```
import Pkg; Pkg.add("Symbolics")

using Symbolics

@variables x
D = Differential(x)

expr = acos(x)

calculated_derivative = expand_derivatives(D(expr))

println("Function:   ", expr)
println("Derivative: ", calculated_derivative)

manual_formula = -1 / sqrt(1 - x^2)

difference = simplify(calculated_derivative - manual_formula)

if isequal(difference, 0)
    println(" Verification Successful.")
else
    println(" Verification Failed.")
```

```
end
```

3 Problem 3 DONE

3.1 Task

Suppose that $f, g : U \to \mathbb{R}$ are continuous in $x_0 \in U$. Prove that if f is also differentiable in x_0 and $f(x_0) = 0$ then the product fg is differentiable in x_0. Illustrate this claim in Julia for the functions $f(x) = x$ and $g(x) = |x|$.

3.2 Solution

> Let $U \subseteq \mathbb{R}$ be an open interval and $f : U \to \mathbb{R}$. We say that f is differentiable at $x \in U$ if the limit
>
> $$\lim_{y \to x} \frac{f(y) - f(x)}{y - x}$$
>
> exists. The value of this limit is denoted by $f'(x)$ and is called a derivative of f.

To prove the product fg is differentiable in x_0, I have to examine the limit of the difference quotient for the product function $(fg)(x)$ at x_0.

$$\lim_{x \to x_0} \frac{(fg)(x) - (fg)(x_0)}{x - x_0}$$

Given the hypothesis that $f(x_0) = 0$, limit can be simplified.

$$\begin{aligned}
\lim_{x \to x_0} \frac{(fg)(x) - (fg)(x_0)}{x - x_0} &= \lim_{x \to x_0} \frac{(fg)(x)}{x - x_0} \\
&= \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} g(x) \\
&= \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} \cdot \lim_{x \to x_0} g(x) \\
&= f'(x_0) g(x_0)
\end{aligned}$$

Since the limit exists, the function product is differentiable at x_0.

```
using Plots

f(x) = x
g(x) = abs(x)
fg(x) = f(x) * g(x)

x = -2:0.1:2
f_x = f.(x)
g_x = g.(x)
fg_x = fg.(x)


plot(x, fg_x,
     title="fg product",
     label="fg(x)",
     xlabel="x", ylabel="fg(x)",
     lw=2)

plot!(x, f_x,
      label="f(x)",
      xlabel="x", ylabel="f(x)",
      lw=2)

plot!(x, g_x,
      label="g(x)",
      xlabel="x", ylabel="g(x)",
      lw=2)

savefig("./figs/p_03.png")
```

4 Problem 4 DONE

4.1 Task

The aim is to prove $(1+\frac{1}{n})^n \to e$ (a) Determine the values of $\log(1)$ $\log'(1)$ and verify the identity $\log'(1) = \lim_{n\to\infty} n\ \log(1+\frac{1}{n})$ (b) Use the definition of the general exponential to prove $(1+\frac{1}{n})^n \to \exp(1)$. (This determines the limit of the sequence in Problem 4 of Problem Set 1)

4.2 Solution

4.2.1 a

$$\frac{d}{dx}f(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

$$\log(1) = 0$$

$$\log'(1) = \frac{d}{dx}(\log(1)) = \lim_{n\to 0}\frac{\log(1+n)-\log(1)}{n}$$

$$\log'(1) = \lim_{n\to 0}\frac{\log(1+n)}{n}$$

$$\log'(1) = \lim_{n\to\infty}\frac{\log(1+\frac{1}{n})}{\frac{1}{n}} \quad \log'(1) = \lim_{n\to\infty} n\log(1+\frac{1}{n})$$

4.2.2 b

$$\log'(1) = \frac{1}{1} = 1$$

$$1 = \lim_{n\to\infty} n\log(1+\frac{1}{n})$$

$$1 = \lim_{n\to\infty}\log((1+\frac{1}{n})^n)$$

$$1 = \log(\lim_{n\to\infty}(1+\frac{1}{n})^n)$$

$$e = \lim_{n\to\infty}(1+\frac{1}{n})^n$$

5 Problem 5 DONE

5.1 Task

Prove by induction (and using the product rule) that, for $1 \leq n \in \mathbb{N}$ the derivative of $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^n$ is given by $f'(x) = nx^{n-1}$.

5.2 Notes

Proposition The statement S1, S2, S3, S4, ... are all true.

Proof. (Induction)

1. Prove that the first statement S1 is true
2. Given any integer k >= 1, prove that the statement Sk => Sk+1 is true.

It follows by mathematical induction that every Sn is true.

5.3 Solution

Proposition. For $1 \leq n \in \mathbb{N} : f : \mathbb{R} \to \mathbb{R}, f(x) = x^n \implies f'(x) = nx^{n-1}$.

Proof.

Base Case. $n = 1$. $f(x) = x^1 = x$. Using the limit definition: $f'(x) = \lim_{h\to 0} \dfrac{(x+h)-x}{h} = \lim_{h\to 0} \dfrac{h}{h} = 1$.

Using the formula: $nx^{n-1} = 1x^{1-1} = 1$.

Base holds.

Induction Hypothesis. Assume statement holds for $n = k : \dfrac{d}{dx}(x^k) = kx^{k-1}$.

Inductive Step. We must show that the statement holds for $n = k+1$. Target: $\dfrac{d}{dx}(x^{k+1}) = (k+1)x^k$.

Rewrite the term: $x^{k+1} = x \cdot x^k$

Apply the Product Rule ($u = x, v = x^k$): $\dfrac{d}{dx}(x^{k+1}) = \dfrac{d}{dx}\left[x \cdot x^k\right]$

$\dfrac{d}{dx}(x \cdot x^k) = \left[\dfrac{d}{dx}x\right] \cdot x^k + x \cdot \left[\dfrac{d}{dx}x^k\right]$

Substitute derivatives (Base Case and Hypothesis): $\dfrac{d}{dx}(x \cdot x^k) = (1) \cdot x^k + x \cdot (kx^{k-1})$

Simplify: $\dfrac{d}{dx}(x^{k+1}) = x^k + kx^{1+k-1}$ $\dfrac{d}{dx}(x^{k+1}) = x^k + kx^k$

Factor: $\dfrac{d}{dx}(x^{k+1}) = (1+k)x^k$ $\dfrac{d}{dx}(x^{k+1}) = (k+1)x^k$

Conclusion. Since S_1 is true and $S_k \implies S_{k+1}$, the proposition is true for all $n \in \mathbb{N}$ by mathematical induction.

6 Problem 6 DONE

6.1 Task

Compute the derivative of the following functions:

1. $f(x) = \sin(x^2)$
2. $g(x) = e^{3x+1}$
3. $f(x) = \ln(5x^2 - x + 4)$
4. $g(x) = (2x^3 - x)^5$
5. $f(x) = \sqrt{1 + e^{2x}}$

6. $g(x) = \cos(\ln\ x)$, $x > 0$

Visualize all functions and their derivatives in Julia.

6.2 Notes

Chain Rule:

$$\frac{d}{dx}\left[f(g(x))\right] = f'(g(x)) \cdot g'(x)$$

Product Rule:

$$\frac{d}{dx}\left[f(x)g(x)\right] = f(x)g'(x) + f'(x)g(x)$$

Derivatives of Logarithmic Functions:

$$\frac{d}{dx} log_a\left[f(x)\right] = \frac{1}{f(x)ln(a)} \cdot f'(x)$$

6.3 Solution

1. $f(x) = \sin(x^2)$
   $f'(x) = 2x\cos(x^2)$ (Chain Rule)
2. $g(x) = e^{3x+1}$
   $g'(x) = 3e^{3x+1}$ (Chain Rule)
3. $f(x) = \ln(5x^2 - x + 4)$
   $f'(x) = \dfrac{10x - 1}{5x^2 - x + 4}$ (Chain Rule)
4. $g(x) = (2x^3 - x)^5$
   $g'(x) = 5(2x^3 - x)^4(6x^2 - 1)$ (Chain Rule)
5. $f(x) = \sqrt{1 + e^{2x}}$
   $f'(x) = e^{2x}(1 + e^{2x})^{-\frac{1}{2}}$ (Chain Rule)
6. $g(x) = \cos(\ln(x))$, $x > 0$
   $g'(x) = \frac{-\sin(ln(x))}{x}$ (Chain Rule)

6.3.1 Code

```
using Plots

const PATH = "./figs/"

# format: (function, derivative, range, title, name)
data = [
    (x -> sin(x^2),          x -> 2x * cos(x^2),                 -3:0.01:3,     "f(x) =
    (x -> exp(3x + 1),       x -> 3 * exp(3x + 1),               -1:0.01:0.5,   "g(x) =
    (x -> log(5x^2 - x + 4), x -> (10x - 1) / (5x^2 - x + 4),    -3:0.01:3,     "f(x) =
    (x -> (2x^3 - x)^5,      x -> 5 * (2x^3 - x)^4 * (6x^2 - 1), -1.1:0.01:1.1, "g(x) =
    (x -> sqrt(1 + exp(2x)), x -> exp(2x) / sqrt(1 + exp(2x)),   -2:0.01:2,     "f(x) =
    (x -> cos(log(x)),       x -> -sin(log(x)) / x,              0.2:0.01:6,    "g(x) =
]

plots = map(data) do (f, df, r, t, n)
    p = plot(r, f, label="f(x)", linewidth=2, title=t, legend=:topleft)
    plot!(p, r, df, label="f'(x)", linewidth=2, title=t, linestyle=:dash)
    savefig(joinpath(PATH, n))
    p
end

final_plot = plot(plots..., layout=(3, 2), size=(1000, 1200), plot_title="Functions and
savefig(joinpath(PATH, "derivatives"))
```

7 Problem 7

7.1 Task

Prove the inequality of arithmetic and geometric means, $1 $\frac{}{2(x+y)\geq}\sqrt{xy}$, $\forall x, y \in [0, \infty)$. (Can be useful for the next exercise!)

7.2 Solution

8 Problem 8 DONE

8.1 Task

Let $(x_n)_{n\geq 0}$ be defined by $x_0 = 1$, $x_{n+1} = \frac{1}{2}(x_n + \frac{2}{x_n})$ $n \in \mathbb{N}$

1. Determine x_1, x_2, and x_3.
2. Compute x_n for large n in Julia, guess the limit, and visualize the convergence.
3. Prove that $x_n \geq \sqrt{2}$ for all $n \geq 1$.
4. Prove that $(x_n)_{n\geq 1}$ is decreasing.
5. Conclude that $(x_n)_{n\in\mathbb{N}}$ converges and determine its limit.

8.2 Solution

- Part (1)

$$
\begin{aligned}
x_0 &= 1 \\
x_1 &= \frac{1}{2}(1 + \frac{2}{1}) = \frac{3}{2} \\
x_2 &= \frac{1}{2}(\frac{3}{2} + \frac{2}{\frac{3}{2}}) = \frac{17}{12} \\
x_3 &= \frac{1}{2}(\frac{17}{12} + \frac{2}{\frac{17}{12}}) = \frac{577}{408}
\end{aligned}
$$

- Part (2)

```
using Plots

term(x_n) = 1/2 * (x_n + (2/x_n))

function main()
    arr = [1.0]

    for i in 1:10-1
        push!(arr, term(last(arr)))
    end


    plot(1:10, arr,
```

```
            title="sqrt(2) approxiamtion",
            label="x_{n+1} = 1/2 * (x_n + (2/x_n))",
            xlabel="x", ylabel="x_n")

    savefig("./figs/p_08.png")
end

main()
```

- Part (3)

$$
\begin{aligned}
x_{n+1} - \sqrt{2} &= \frac{1}{2}(x_n + \frac{2}{x_n}) - \sqrt{2} \\
&= \frac{x_n}{2} + \frac{1}{x_n} - \sqrt{2} \\
&= \frac{x_n^2}{2x_n} + \frac{2}{2x_n} - \frac{2x_n\sqrt{2}}{2x_n} \\
&= \frac{x_n^2 + 2 - 2x_n\sqrt{2}}{2x_n} \\
&= \frac{x_n^2 - 2\sqrt{2}x_n + 2}{2x_n} \\
&= \frac{(x_n - \sqrt{2})^2}{2x_n}
\end{aligned}
$$

Since $(x_n - \sqrt{2})^2 \geq 0$ and $2x_n > 0$ (as $x_n > 0$), $\frac{(x_n - \sqrt{2})^2}{2x_n} \geq 0$.

- Part (4)

$$
\begin{aligned}
\frac{x_{n+1}}{x_n} &= \frac{\frac{1}{2}(x_n + \frac{2}{x_n})}{x_n} \\
&= \frac{x_n + \frac{2}{x_n}}{2x_n} \\
&= \frac{x_n^2 + 2}{2x_n^2}
\end{aligned}
$$

Since $x_n \geq \sqrt{2}$ (as proved in part (3)):

- Case 1: $x_n = \sqrt{2}$

$$\begin{aligned}\frac{(\sqrt{2})^2+2}{2(\sqrt{2})^2} &= \frac{2+2}{2\cdot 2}\\ &= \frac{4}{4}\\ &= 1\end{aligned}$$

- Case 2: $x_n > \sqrt{2}$
  As for $x_n > \sqrt{2}$, $x_n^2 + 2 < x_n^2 + x_n^2$. Thus,

$$\frac{x_n^2+2}{2x_n^2} < 1$$

  Hence, $(x_n)_{n\geq 1}$ is decreasing.

- Part (5)

  Since $(x_n)_{n\geq 1}$ is monotone decreasing and bounded from below ($x_n \geq \sqrt{2}$), by the Monotone Convergence Theorem, the sequence $(x_n)_{n\geq 1}$ is convergent and has a limit $L \geq \sqrt{2}$.

  I find the limit by taking a limit of both sides:

$$\begin{aligned}\lim_{x_{n+1}\to L} x_{n+1} &= \lim_{x_n\to L} \frac{1}{2}(x_n + \frac{2}{x_n})\\ L &= \frac{1}{2}(L + \frac{2}{L})\\ 2L &= L + \frac{2}{L}\\ L &= \frac{2}{L}\\ L^2 &= 2\\ L &= \sqrt{2}\end{aligned}$$

9 Problem 9 DONE

9.1 Task

Consider $f : \mathbb{R} \to \mathbb{R}$ $f(x) = \cos(x)$ and the sequence $x_0 = 2$, $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ $n \in \mathbb{N}$ Implement the sequence in Julia and compare with the bisection method in the interval [0, 2].

9.2 Solution

```
using Plots

# ensure output directory exists
mkpath("./figs")

# define functions
f(x) = cos(x)
df(x) = -sin(x)
newton_step(x) = x - f(x) / df(x)

# shared settings
iterations = 5

# --- part 1: Newton's Method ---
x0 = 2.0
n_history = Float64[x0]

for i in 1:iterations
    current_x = last(n_history)
    next_x = newton_step(current_x)

    push!(n_history, next_x)
    println("Newton Iter $i: $next_x")
end

# --- part 2: Bisection Method ---
a, b = 0.0, 2.0
b_history = Float64[]

curr_a, curr_b = a, b
```

```
for i in 1:iterations
    global curr_a, curr_b

    c = (curr_a + curr_b) / 2
    push!(b_history, c)

    if f(c) * f(curr_a) < 0
        curr_b = c
    else
        curr_a = c
    end
    println("Bisection Iter $i: $c" )
end


# plotting
indices_n = 0:(length(n_history) - 1)
indices_b = 1:length(b_history)

plot(title="Convergence Comparison: Newton vs Bisection",
     xlabel="Iteration", ylabel="Value", legend=:topright)

# plot Newton
plot!(indices_n, n_history,
      label="Newton (x_n)",
      linewidth=2, marker=:circle, color=:blue)

# plot Bisection
plot!(indices_b, b_history,
      label="Bisection (midpoints)",
      linewidth=2, marker=:square, color=:green, linestyle=:dash)

# reference Line
hline!([pi/2], label="True Root (/2)", color=:red, linestyle=:dot)

savefig("./figs/p_9")
```